

Principles of Inheritance and Variation

Chapter 4 · All questions with full solutions



Part 3 · Questions and Solutions

Q

1

Advantages
of pea plant
for Mendel's
experiments



Mention the advantages of selecting pea plant for experiment by Mendel.

Given

Mendel selected pea plants to carry out his study on the inheritance of characters from parents to offspring.

Reasons pea plants were chosen: He selected a pea plant because of the following features.

- Peas have many visible contrasting characters such as tall/dwarf plants, round/wrinkled seeds, green/yellow pod, purple/white flowers, etc.

- Peas have bisexual flowers and therefore undergo self pollination easily.

Thus, pea plants produce offsprings with same traits generation after generation.

- In pea plants, cross pollination can be easily achieved by emasculation in which the stamen of the flower is removed without affecting the pistil.

- Pea plants have a short life span and produce many seeds in one generation.

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2

Key genetics
term pairs



Differentiate between the following -

Part (a)

Dominance and Recessive

	Dominance	Recessive
1.	A dominant factor or allele expresses itself in the presence or absence of a recessive trait.	A recessive trait is able to express itself only in the absence of a dominant trait.
2.	For example, tall plant, round seed, violet flower, etc. are dominant characters in a pea plant.	For example, dwarf plant, wrinkled seed, white flower, etc. are recessive traits in a pea plant.

Part (b)

Homozygous and Heterozygous

	Homozygous	Heterozygous
1.	It contains two similar alleles for a particular trait.	It contains two different alleles for a particular trait.
2.	Genotype for homozygous possess either dominant or recessive, but never both the alleles. For example, RR or rr	Genotype for heterozygous possess both dominant and recessive alleles. For example, Rr
3.	It produces only one type of gamete.	It produces two different kinds of gametes.

Part (c)

Monohybrid and Dihybrid.

	Monohybrid	Dihybrid
1.	Monohybrid involves cross between parents, which differs in only one pair of contrasting characters.	Dihybrid involves cross between parents, which differs in two pairs of contrasting characters.

2.

For example, the cross between tall and dwarf pea plant is a monohybrid cross.

For example, the cross between pea plants having yellow wrinkled seed with those having green round seeds is a dihybrid cross.

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3

Gametes from a tetra-heterozygous organism

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A diploid organism is heterozygous for 4 loci, how many types of gametes can be produced?

Locus

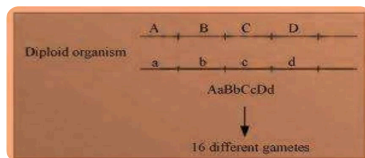
Locus is a fixed position on a chromosome, occupied by one or more genes.

Substitute

Heterozygous organisms contain different alleles for an allelic pair.

So, a diploid organism that is heterozygous at four loci will have four different contrasting characters at four different loci.

For example, if an organism is heterozygous at four loci with four characters, say Aa, Bb, Cc, Dd, then during meiosis, it will segregate to form 16 separate gametes.



Conclusion

If the genes are not linked, the diploid organism will produce 16 different gametes. However, if the genes are linked, they are inherited together during meiosis, so the number of different gametes produced will be lower.

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4

Law of Dominance via monohybrid cross

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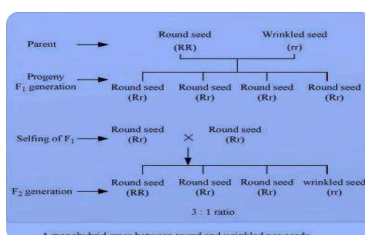
Explain the Law of Dominance using a monohybrid cross.

Law of Dominance

Mendel's law of dominance states that a dominant allele expresses itself in a monohybrid cross and suppresses the expression of recessive allele. However, this recessive allele for a character is not lost and remains hidden or masked in the progenies of F_1 generation and reappears in the next generation.

Conclusion

For example, when pea plants with round seeds (RR) are crossed with plants with wrinkled seeds (rr), all seeds in F_1 generation were found to be round (Rr). When these round seeds were self-fertilized, both the round and wrinkled seeds appeared in F_2 generation in 3: 1 ratio. Hence, in F_1 generation, the dominant character (round seeds) appeared and the recessive character (wrinkled seeds) got suppressed, which reappeared in F_2 generation.



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5

Test cross
definition
and design



Define and design a test-cross.

Test cross

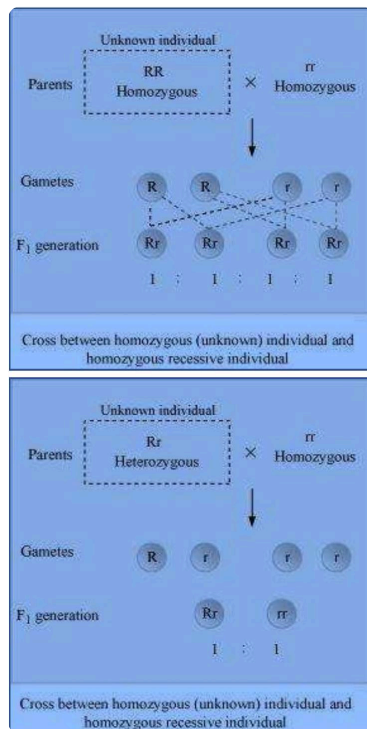
Test cross is a cross between an organism with unknown genotype and a recessive parent.

It is used to determine whether the individual is homozygous or heterozygous for a trait.

Conclusion

If the progenies produced by a test cross show 50% dominant trait and 50% recessive trait, then the unknown individual is heterozygous for a trait.

On the other hand, if all the progeny produced show dominant trait, then the unknown individual is homozygous for a trait.



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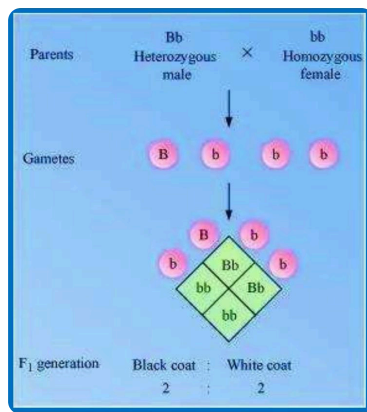
Punnett
square for a
monohybrid
cross



Using a Punnett Square, workout the distribution of phenotypic features in the first filial generation after a cross between a homozygous female and a heterozygous male for a single locus.

Conclusion

In guinea pigs, heterozygous male with black coat colour (Bb) is crossed with the female having white coat colour (bb). The male will produce two types of gametes, B and b, while the female will produce only one kind of gamete, b. The genotypic and phenotypic ratio in the progenies of F₁ generation will be same i.e., 1:1.



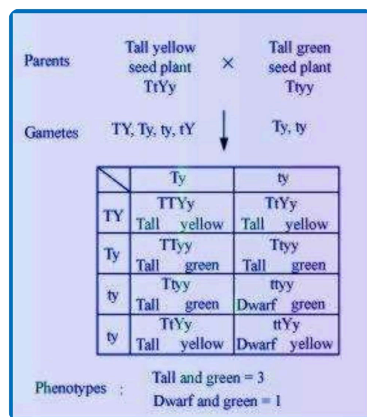
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7

Dihybrid cross phenotype ratio

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When a cross is made between tall plant with yellow seeds (TtYy) and tall plant with green seed (Tt yy), what proportions of phenotype in the offspring could be expected to be



Conclusion

A cross between tall plant with yellow seeds and tall plant with green seeds will produce the following types and proportions of phenotype in the offspring: (a) tall and green plants in a proportion of three out of eight (3/8) (b) dwarf and green plants in a proportion of one out of eight (1/8).

Part (a)

tall and green.

Conclusion

three tall and green plants

Part (b)

dwarf and green.

Conclusion

one dwarf and green plant

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8

Linked genes in a dihybrid cross



Two heterozygous parents are crossed. If the two loci are linked what would be the distribution of phenotypic features in F₁ generation for a dihybrid cross?

Linkage

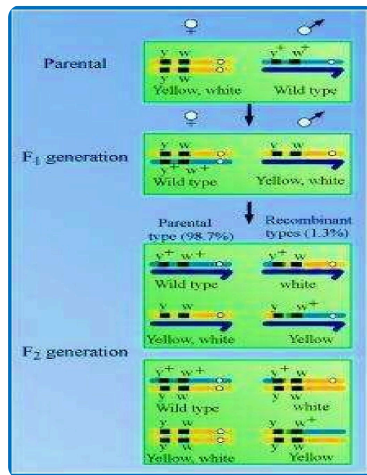
Linkage is the coexistence of two or more genes on the same chromosome.

Genes that lie close together on the same chromosome are inherited together, and are called linked genes.

Conclusion

For example, a cross between a yellow-bodied, white-eyed Drosophila and a wild-type Drosophila produces wild-type and yellow-white progenies, because the yellow-body

and white-eye genes are linked and so inherited together.



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9

Morgan's
contribution
to genetics



Briefly mention the contribution of T.H. Morgan in genetics.

Conclusion

Morgan's work is based on fruit flies (*Drosophila melanogaster*).

He experimentally verified the chromosomal theory of inheritance (first proposed by Sutton and Boveri) and, through dihybrid crosses in *Drosophila* studying sex-linked genes, found that genes situated on the same chromosome do not segregate independently.

He coined the term linkage to describe this physical association of genes on a chromosome, and the term recombination for the generation of non-parental gene combinations.

His experiments have also proved that tightly linked genes show very low recombination while loosely linked genes show higher recombination.

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10

Pedigree
analysis



What is pedigree analysis? Suggest how such an analysis, can be useful.

Pedigree analysis

Pedigree analysis is a record of how a trait appears across several generations of a family.

It is based on the fact that certain characteristic features are heritable in a family, for example, eye colour, skin colour, hair form and colour, and other facial characteristics.

Conclusion

Genetic disorders, such as Mendelian disorders, are also inherited in a family from one generation to the next.

So, pedigree analysis can be used to study specific traits or disorders across generations and trace their pattern of inheritance.

In this analysis, the inheritance of a trait is represented as a tree, called a family tree.

Genetic counsellors use a pedigree chart to analyse various traits and diseases in a family and predict their inheritance patterns.

It is useful in preventing hemophilia, sickle cell anemia, and other genetic disorders in future generations.

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11

Sex
determination
in humans

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How is sex determined in human beings?

Conclusion

Human beings exhibit male heterogamety.

In humans, males (XY) produce two different types of gametes, X and Y.

The human female (XX) produces only one type of gamete, containing an X chromosome.

The sex of the baby is determined by the type of male gamete that fuses with the female gamete.

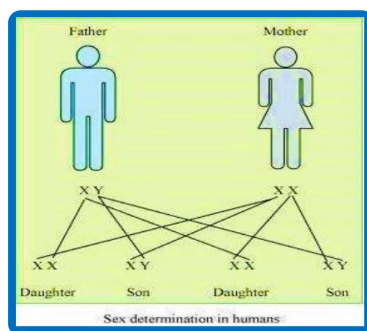
If the fertilizing sperm contains an X chromosome, the baby produced will be a girl.

If it contains a Y chromosome, the baby produced will be a boy.

So, it is a matter of chance that determines the sex of a baby.

There is an equal probability of the fertilizing sperm carrying an X or Y chromosome.

Thus, it is the genetic makeup of the sperm that determines the sex of the baby.



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12

ABO blood
group
inheritance

A child has blood group O.

If the father has blood group A and mother blood group B, work out the genotypes of the parents and the possible genotypes of the other offsprings.

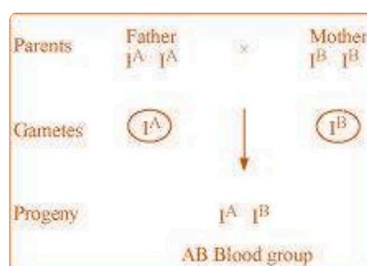
ABO blood
group
alleles

The blood group characteristic in humans is controlled by three set of alleles, namely, I^A , I^B , and i . The alleles, I^A and I^B , are equally dominant whereas allele, i , is recessive to the other alleles. The individuals with genotype, $I^A I^A$ and $I^A i$, have blood group A whereas the individuals with genotype, $I^B I^B$ and $I^B i$, have blood group B. The persons with genotype $I^A I^B$ have blood group AB while those with blood group O have genotype ii .

Substitute

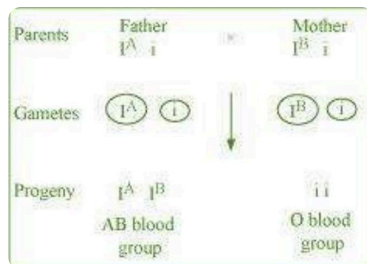
So, if the father has blood group A and the mother has blood group B, the possible genotypes of the parents will be: Father - $I^A I^A$ or $I^A i$; Mother - $I^B I^B$ or $I^B i$.

A cross between homozygous parents will produce progeny with AB blood group.



Conclusion

A cross between heterozygous parents will produce progenies with AB blood group ($I^A I^B$) and O blood group (ii). Since the child in this problem has blood group O (genotype ii), both parents must carry the recessive allele i - the father's genotype is $I^A i$ and the mother's genotype is $I^B i$. Besides the child with blood group O (ii), the other possible offspring genotypes from this cross are $I^A I^B$ (AB blood group), $I^A i$ (A blood group), and $I^B i$ (B blood group).

**Q****13**

Co-
dominance
and
incomplete
dominance

**Explain the following terms with example**

Part (a)

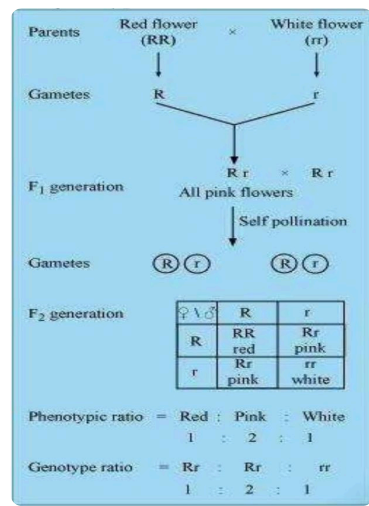
Co-dominance**Co-
dominance**

Co-dominance is the phenomenon in which both the alleles of a contrasting character are expressed in heterozygous condition. Both the alleles of a gene are equally dominant. ABO blood group in human beings is an example of codominance. The blood group character is controlled by three sets of alleles, namely, I^A , I^B , and i . The alleles, I^A and I^B , are equally dominant and are said to be co-dominant as they are expressed in AB blood group. Both these alleles do not interfere with the expression of each other and produce their respective antigens. Hence, AB blood group is an example of co-dominance.

Part (b)

Incomplete dominance**Incomplete
dominance**

Incomplete dominance is a phenomenon in which one allele shows incomplete dominance over the other member of the allelic pair for a character. For example, a monohybrid cross between the plants having red flowers and white flowers in *Antirrhinum* species will result in all pink flower plants in F_1 generation. The progeny obtained in F_1 generation does not resemble either of the parents and exhibits intermediate characteristics. This is because the dominant allele, R , is partially dominant over the other allele, r . Therefore, the recessive allele, r , also gets expressed in the F_1 generation resulting in the production of intermediate pink flowering progenies with Rr genotype.



Q

14

Point mutation



What is point mutation? Give one example.

Point mutation

Point mutation is a change in a single base pair of DNA by substitution of one nitrogenous base for another (deletion or insertion of a base instead causes a frame-shift mutation).

Conclusion

An example of point mutation is sickle cell anaemia.

It involves a single base substitution (from GAG to GUG) at the sixth codon of the beta-globin gene, located on chromosome 11, causing glutamic acid to be replaced by valine at the sixth position of the beta-globin chain of haemoglobin.

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15

Chromosomal theory of inheritance

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Who had proposed the chromosomal theory of the inheritance?

Conclusion

Sutton and Boveri proposed the chromosomal theory of inheritance in 1902. They linked the inheritance of traits to the chromosomes.

Q

16

Autosomal genetic disorders



Mention any two autosomal genetic disorders with their symptoms.

Given

Two autosomal genetic disorders are as follows.

Sickle cell Anaemia (a Mendelian, single-gene disorder): It is an autosomal linked recessive disorder, which is caused by point mutation in the beta-globin chain of haemoglobin pigment of the blood. The disease is characterized by sickle shaped red blood cells, which are formed due to the mutant haemoglobin molecule. The disease is controlled by Hb^A and Hb^S allele. The homozygous individuals with genotype, Hb^S Hb^S, show the symptoms of this disease while the heterozygous individuals with genotype, Hb^A Hb^S, are not affected. However, they act as carriers of the disease. Symptoms: Rapid heart rate, breathlessness, delayed growth and puberty, jaundice, weakness, fever, excessive thirst, chest pain, and decreased fertility are the major

symptoms of this disease.

Down's syndrome (a chromosomal disorder): It is an autosomal disorder that is caused by the trisomy of chromosome 21.

Symptoms: The individual is short statured with a small round head, furrowed tongue, and partially open mouth.

The palm is broad with a characteristic palm crease. Physical, psychomotor, and mental development is retarded.